

Project Update 2

ECON 345

Due: 11:59PM September 26, 2026

Instructions (Do Not Skip)

Completing a thoughtful research project doesn't happen overnight. Accordingly, we have designed 6 project update assignments that will guide you through the research process. These project updates build upon each other, meaning that you will need to complete Project Update #2 in order to successfully accomplish Project Update #3. Once you submit a project update, your project buddy and economist advisor will provide feedback on your work. If you thoughtfully complete all 6 project updates and routinely incorporate feedback from your buddy and advisor, you will be on track for a solid final research presentation and paper.

The goal of Project Update #2 is for you to:

- Begin transforming your data
- Create and interpret summary statistics
- Analyze your results

You will not receive credit for work that is missing or incomplete without an explanation of what you tried and how you're struggling. For any parts you are unable to run in RStudio and produce the output for, please include the code (even if it does not work) and briefly explain what you did to attempt the task, and what problems arose (were there specific error messages?). In order to receive partial credit, you must include the code you attempted.

If you run into any difficulties while trying to complete this assignment, consider the following resources:

- Email your adviser (w. econ345.spelman@gmail.com cc'd) with questions on research design (example: what variables would it be interesting to explore the relationship of in order to support my hypothesis?). They want to hear from you often!
- For technical help (example: my code is breaking or the formatting of my chart looks off, but I haven't been able to figure out why!), post a question on Canvas, email econ345.spelman@gmail.com, or come to office hours. Atlanta Fed staff are eager to help!

Instructions for completing this assignment (do not skip)

For this assignment, you will need to submit i) your RMarkdown (RMD) file and ii) the resulting PDF or Word document.

RMD File

The RMD file is where you will compose your project's production code alongside written interpretation in order to easily and clearly communicate your research. Recall, in the last assignment you had to copy and paste screen-grabs of your data from RStudio into a Word doc. This method is really inefficient given that you'll likely need to make many revisions to a given product—remember: the research process is very iterative! RMD helps to cut out the middleman so-to-speak. In this file, you should organize pieces of your code (from this week's section of your lab notebook i.e., R script) into well-labeled chunks throughout your RMD file. Then either above or below these chunks, you can write out your response/interpretation. Ideally, you should knit your RMD to PDF so we can easily view your results and provide feedback. Alternatively, you can knit

to Word. If you cannot knit, submit screenshots of your output and your interpretation in a word document.

Part A - Data Transformations (20 points)

In last week's update, you loaded your data into R and identified variables of interest.

1. List the variables that need to be transformed and the accompanying transformations (ie, if you have a continuous variable for income and you need to transform it into categorical variables of "low", "middle", or "high" income) (10 points)
2. Do the appropriate transformations and assign them to a new column in the dataframe. Provide the code below that does this. (10 points)

Part B - Compute summary statistics (60 points)

The previous two exercises are aimed at developing your conceptual understanding of your research question, establishing how the data available to you may lend some insights into your research, and anticipating what you may find. Finally, you're ready to start digging into your data. To this end:

3. **Compute summary statistics for your key (i.e., dependent and independent) variables. (30 point)**
 - If your variables are continuous, consider mean, median, sd, min, max, P25, P75, and N.
 - If your variables are categorical, consider frequency counts and sample shares/proportions.

A straightforward way to create summary tables in R is by using the vtable package, though this is not required. See the file "summary_tables_with_vtable_package.pdf" for an example.

4. **Provide an interpretation of the summary statistics for each of your key variables (30 points)** Consider:
 - How do these summary statistics compare to your expectations discussed above?
 - Was there anything that surprised you?
 - Do these summary statistics reveal any data transformation needs?

Example: Your summary statistics are on how many miles per gallon different types of cars get: $N = 32$, mean = 20.1, stdev = 6.0, min = 10.4, 25th pct = 15.4, 50th pct = 19.2, 75th pct = 22.8, max = 33.9.

Do not do the following: *"There are 32 observations, the mean is 20.1, the standard deviation is 6.0, the minimum is 10.4, the 25th percentile is 15.4, the median is 19.2, the 75th percentile is 22.8, and the maximum value is 33.9." This is not an interpretation because it is simply repeating the data from the table in sentence form.*

Instead, focus on telling a story that gives meaning to the values. For example: *"This data set contains 32 observations which each represent a different type of vehicle. The average miles per gallon across all vehicles is 20.1, which seems reasonable. The mean of 20.1 is slightly larger than the median of 19.2, indicating the data is slightly right skewed, although the skew is not particularly strong. The miles per gallon across all cars in the data set ranges from a minimum value of 10.4 to a maximum value of 33.9. This is in line with what would be expected in a sample of different vehicle types, and therefore I was not surprised by these results. If this had been a data set of only electric vehicles, I would've been surprised to see such a small minimum value of 10 MPG since I'd think they would get better mileage. I do not believe I need to make any data transformations since these values are small, easy to work with, and there are no missing values."*

Part C - Follow-up With Your Adviser (20 points)

5. Email your adviser with the knitted output (i.e., the PDF or Word document that results from your RMD) and suggest some meeting times for next week. Do not send them your R script file or your

RMD, but do submit these on canvas. In your email, consider and **propose solutions** to the following questions with your adviser.

- What have you learned about your variables? (5 points)
- Does this new information suggest that you need to transform these variable to better execute your analysis? How would you do so? (5 points)
- What have you learned about the relationship of interest? (5 points)
- How does this fit with your expectations? (5 points)

To receive credit for this, you must cc econ345.spelman@gmail.com.